

Symbolic Regression for Finding Green's Functions

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Abstract

Green's functions are solutions to linear operators such that the source is a delta function. If the delta source is replaced by a general inhomogeneous source, then the general solution is the convolution of the source with the linear operators Green's function. Many Green's functions are known, but finding one for a general operator of physical interest is often difficult and requires non-trivial amounts of effort. In this paper I use the well known solver Dedalus to generate solutions for well known and less well known linear differential equations under near delta forcing. I then use symbolic regression to find the Green's function for the numerically solved differential equations. For the Laplacian operator I compare in two and three dimensions and find the Green's functions to approximately be $\ln(r)$ and $\frac{1}{r}$ as expected. I then performed the method on the Ornstein-Uhlenbeck operator and got a result that indicates the importance of simulation on the final result.

1 Methodology

There are two parts for performing what I did: first solve the differential equation with a point source using Dedalus[1] then convert the solution into an appropriate distance dependent form from the source and use pysr [2] to perform symbolic regression. The individual details for the simulation and the symbolic regression can be found in the uploaded code.

2 Laplacian

The Green's function of the Laplace operator, ∇^2 , is known to depend on the dimension of your system. In two dimensions $G(r - r') = \ln(r - r')$ and in three dimensions

$$G(r - r') = \frac{1}{r - r'}$$

. Using my method I found the Green's functions to most likely be

$$\ln\left(\frac{r - r'}{5.2}\right) * 0.022$$

and

$$-0.033(r - r')^{-1.1}$$

for 2D and 3D respectively. Interesting and importantly, for the 2D Laplacian I simulated within a box with distance 5 from the center and the solution finds that very well.

3 Ornstein-Uhlenbeck

The Ornstein-Uhlenbeck process describes a Brownian particle diffusing in a harmonic potential. The linear operator I solved for is given by $\partial_t f(x, t) - \nabla^2 f(x, t) + x \cdot \nabla f(x, t) = 0$. I found the log of the Green's function to be

$$-1.19 + \frac{0.835 - r}{\left(t + \frac{1}{r} \cdot 4.55\right) (e^t - 0.625)}$$

for one simulation and regression. The expected result is

$$\ln u = -\frac{r^2}{2(2 - e^{-2t})} - \frac{1}{2} \ln(2\pi(2 - e^{-2t}))$$

Clearly different without the same form. The simulation parameters very clearly matter. I took the log of the result because I knew the answer would be an exponential.

4 Conclusion

This simple methodology works well enough, but is highly simulation dependent. If one knows the form of the solution and the terms it must contain then one can even massage the simulation and the regression into giving the "correct" answer. But, if one is fully unaware, or worse, naive in the properties of PDE solving and how the PDE itself and it's Green's function interacts with boundary conditions, then the guess may as well not exist.

References

- [1] Keaton J. Burns, Geoffrey M. Vasil, Jeffrey S. Oishi, Daniel Lecoanet, and Benjamin P. Brown. Dedalus: A flexible framework for numerical simulations with spectral methods. *Physical Review Research*, 2(2):023068, April 2020.
- [2] Miles Cranmer. Interpretable Machine Learning for Science with PySR and SymbolicRegression.jl, May 2023. arXiv:2305.01582 [astro-ph.IM].